MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 11 · ALGEBRA II · TOPIC 6

Lesson 6-5

Properties of Logarithms

$$\log_b(MN) = \log_b M + \log_b N$$

Journey of 6 stations, 60 minutes

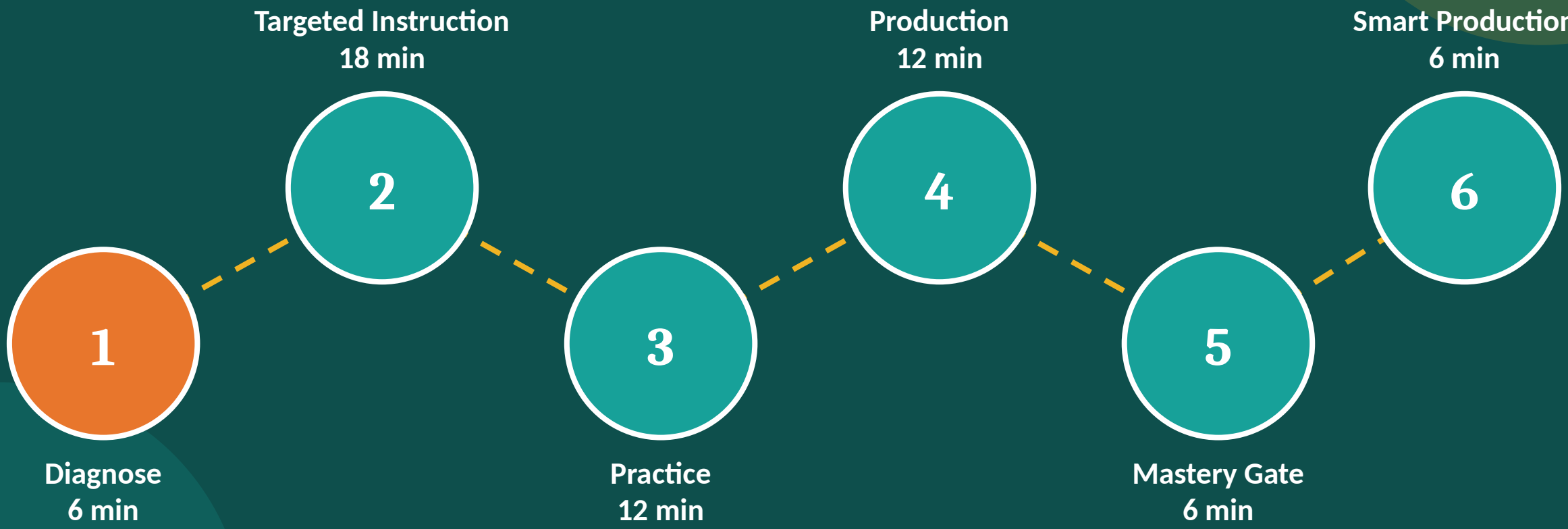
Exponential and Logarithmic Functions

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Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Apply properties of logarithms to rewrite logarithmic expressions and evaluate logarithms.
2. Use the change of base formula to rewrite and evaluate logarithmic expressions.

ESSENTIAL QUESTION

How can the properties of logarithms, including the change of base formula, be applied to simplify and evaluate logarithmic expressions?

MATHEMATICAL PRACTICES MP.2 · MP.7

VOCABULARY

- change of base formula
- logarithmic properties

STANDARDS

HSA.SSE.A.2 Use the structure of an expression to identify ways to rewrite it.

HSF.BF.A.2 Write arithmetic and geometric sequences both recursively and with an explicit formula, use them to model situations, and translate between the two forms.

HSF.LE.A.4 For exponential models, express as a logarithm the solution to $ab^{ct} = d$ where a, c, and d are numbers and the base b is 2, 10, or e; evaluate the logarithm using technology.

STATION 1 · DIAGNOSE · 6 MIN

Start here: three quick checks

1

Evaluate $\log_2 32$

2

Write $2^3 \cdot 2^4$ as one power

3

Evaluate $\log_{10} 0.01$

Done when

All three answers are written without a calculator, and you can say in one sentence what happens to the exponents when you multiply powers with the same base.

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Products and quotients become sums and differences

$$\log_b(MN) = \log_b M + \log_b N \qquad \log_b \frac{M}{N} = \log_b M - \log_b N$$

EXAMPLE A

Expand $\log_2(8x)$

$= \log_2 8 + \log_2 x$

$= 3 + \log_2 x$

EXAMPLE B

Expand $\log_5 \frac{25}{y}$

$= \log_5 25 - \log_5 y$

$= 2 - \log_5 y$

EXAMPLE C

Condense $\log 4 + \log 25$

$= \log(4 \cdot 25)$

$= \log 100 = 2$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Powers come to the front

$$\log_b M^p = p \log_b M \qquad \log_b \sqrt{M} = \frac{1}{2} \log_b M$$

EXAMPLE A

$$\text{Expand } \log_5 \frac{x^2\sqrt{y}}{z^3}$$

$$= \log_5 x^2 + \log_5 y^{1/2} - \log_5 z^3$$

$$= 2\log_5 x + \tfrac{1}{2}\log_5 y - 3\log_5 z$$

EXAMPLE B

$$\text{Condense } 3\log_2 x - \log_2 y$$

$$= \log_2 x^3 - \log_2 y$$

$$= \log_2 \frac{x^3}{y}$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

The change of base formula

$$\log_b x = \frac{\log_a x}{\log_a b} \qquad \log_b x = \frac{\ln x}{\ln b} = \frac{\log x}{\log b}$$

EXAMPLE A

Evaluate $\log_5 40$

$$= \frac{\ln 40}{\ln 5}$$

$$\approx 2.292 \text{ (calculator)}$$

EXAMPLE B

Evaluate exactly $\log_4 8$

$$= \frac{\log_2 8}{\log_2 4}$$

$$= \frac{3}{2}$$

EXAMPLE C

Solve $3 \cdot 2^t = 100$

$$2^t = \frac{100}{3}$$

$$t = \log_2 \frac{100}{3}$$

$$= \frac{\ln(100/3)}{\ln 2} \approx 5.059$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Logarithms and geometric sequences

$$a_n = a_1\, r^{\,n-1} \qquad r^{\,n-1} = \frac{a_n}{a_1} \;\Rightarrow\; n-1 = \log_r \frac{a_n}{a_1}$$

EXAMPLE A

$a_1 = 3,\ a_{n+1} = 2a_n$
Which term is 768?

$$a_n = 3 \cdot 2^{\,n-1} = 768$$

$$2^{\,n-1} = 256 \Rightarrow n-1 = 8$$

$$n = 9$$

EXAMPLE B

$a_n = 5(1.5)^{n-1}$. First term above 100?

$$1.5^{\,n-1} > 20$$

$$n - 1 > \frac{\ln 20}{\ln 1.5} \approx 7.39$$

$$n = 9 \quad (a_9 \approx 128.1)$$

STATION 3 · PRACTICE · 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE
Secure the method

Expand $\log_3(9x^2)$

Evaluate $\log_6 10$ to 3 decimals

Done when: each property is named beside the step that uses it, and the decimal is rounded only at the last step.

APPLY
Use it in context

$\log_b 2 = 0.3,\ \log_b 3 = 0.48$
Find $\log_b 18$

$a_1 = 2,\ r = 3$
Which term is 486?

Done when: 18 is split into factors before any property is used, and the term number is a whole number checked in the formula.

INVESTIGATE
Find out why

Is $\log(M+N) = \log M + \log N$?

Show why $\log_b M^p = p \log_b M$

Done when: you give a counter-example with numbers for the first, and a reason that starts from writing M as a power of b for the second.

STATION 4 · PRODUCTION · 12 MIN

Production: Red Sea tourism growth

THE MODEL

A model for yearly visitors to a Red Sea destination on the Jeddah coast: 20 000 visitors in year 1, growing by 20 percent each year. Model values only, not a forecast. Each visitor spends a model 1 500 SAR. Planners want to know when the model first passes 100 000 visitors.

$$a_n = 20\,000\,(1.2)^{n-1} \quad \text{(visitors in year } n\text{)}$$

$$R_n = 1500\,a_n \quad \text{(SAR)}$$

1. Write the first four terms and the recursive rule, then check the fourth term with the formula.
2. Use a logarithm and the change of base formula to find the first year with more than 100 000 visitors.
3. Find the model revenue in that year, in millions of SAR.

DONE WHEN

- The terms are 20 000, 24 000, 28 800 and 34 560, and the recursive rule says multiply by 1.2.
- The exact expression for the year number is written before any decimal.
- The year is justified by testing the visitors in the year before and the year itself.
- The revenue is about 154.8 million SAR, written with units.

Hint for solving: divide by 20 000 first, then ask which power of 1.2 gives 5, and use ln on the top and the bottom.

STATION 5 · MASTERY GATE · 6 MIN

Mastery Gate: three questions to pass

1

Condense $2\log_3 x + \log_3 5$

2

Evaluate $\log_7 50$ to 3 decimals

3

$\log_b 2 = 0.3,\ \log_b 5 = 0.7$
Find $\log_b 40$

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 · SMART PRODUCTION · 6 MIN

Smart Production: make one, then check it

CREATE

1. Write one change of base question with a base that is not on the calculator, and one expand-or-condense question with three terms.
2. Write a full solution on the back, naming the property beside every step.
3. Swap with a partner and solve theirs.

Done when: both questions have one correct answer, and your solution names a property or the change of base formula at every step.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

$$\text{Expand } \log_2 \frac{5x^3}{y}$$

$$\text{Evaluate } \log_3 20 \text{ to 3 decimals}$$

Next lesson: 6-6 Exponential and Logarithmic Equations and Inequalities.

Exam connections

SAT

If $2^a = 5$ and $2^b = 3$, what is 2^{2a-b}?

STEPS

1 $$2^{2a-b} = \frac{(2^a)^2}{2^b}$$

2 $$= \frac{5^2}{3}$$

3 $$= \frac{25}{3}$$

GAT

Which quantity is greater?

A: $\log_{10} 25 + \log_{10} 4$ B: 2

STEPS

1 $$\log_{10} 25 + \log_{10} 4 = \log_{10} 100$$

2 $$\text{A} = 2 = \text{B}$$

3 they are equal

SAAT

If $\log_2 3 = a$, then $\log_4 12 = ?$

STEPS

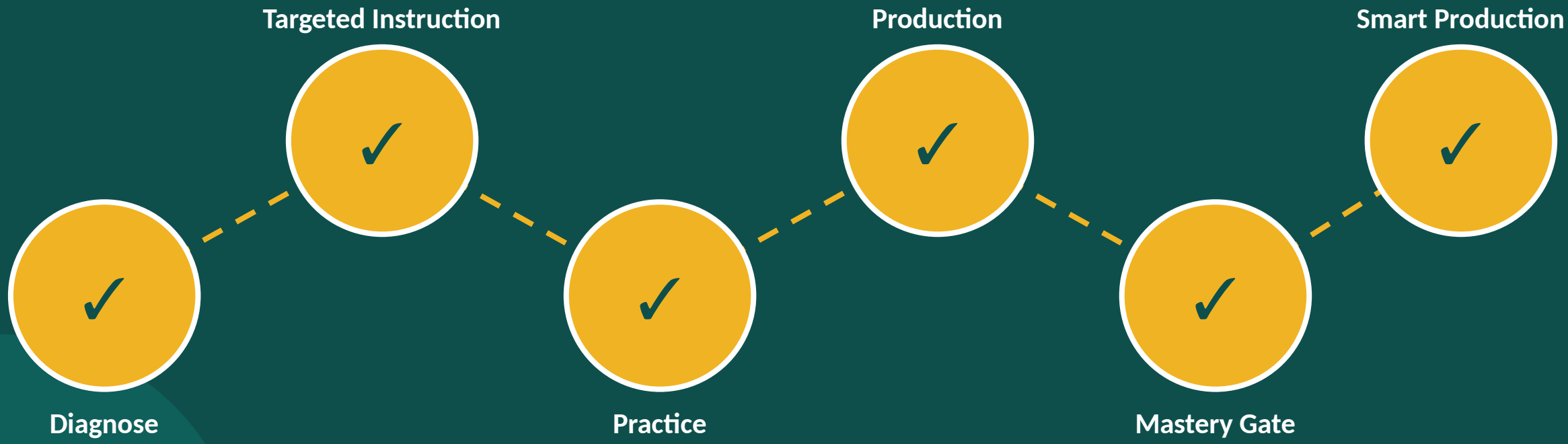
1 $$\log_4 12 = \frac{\log_2 12}{\log_2 4}$$

2 $$= \frac{\log_2 3 + \log_2 4}{2}$$

3 $$= \frac{a+2}{2}$$

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Journey complete



Products become sums, quotients become differences, powers come to the front, and the change of base formula turns any logarithm into a ratio you can evaluate.